

Effects of antiplatelet components of tomato extract on platelet function in vitro and ex vivo: a time-course cannulation study in healthy humans.

BACKGROUND: Natural antithrombotic agents that influence platelet function are of potential interest for primary prevention of cardiovascular disease. Previous reports showed that tomato extracts inhibit platelet aggregation in vitro, but little is known of the active components, their mode of action, or their efficacy in vivo. **OBJECTIVE:** The objectives of the study were to examine the antiplatelet activity of specific tomato components by in vitro experimentation and to establish their ex vivo efficacy in healthy humans. **DESIGN:** The mechanisms of action of antiplatelet components isolated from tomato extracts were examined in vitro. A 7-h time-course study was carried out in cannulated human subjects (n = 23) to determine the ex vivo efficacy of a supplement drink containing tomato extract and the onset and duration of antiplatelet effects. **RESULTS:** The inhibition of ADP-, collagen-, thrombin-, and arachidonate-mediated platelet aggregation by tomato extract components appears to be linked to the inhibition of glycoprotein IIb/IIIa and platelet secretory mechanisms. We found a significant inhibition of baseline platelet function, from 2.9 +/- 1.4% (optimal ADP concentrations; P = 0.03) to 20.0 +/- 4.9% (suboptimal ADP concentrations; P < 0.001), 3 h after supplementation with a dose of tomato extract equivalent to 6 tomatoes. The observed effects persisted for >12 h. Coagulation variables were not affected. **CONCLUSIONS:** The ingestion of tomato components with in vitro antiplatelet activity significantly affects ex vivo platelet function. The reported cardioprotective effects of tomatoes are potentially linked to a modulation of platelet function.